

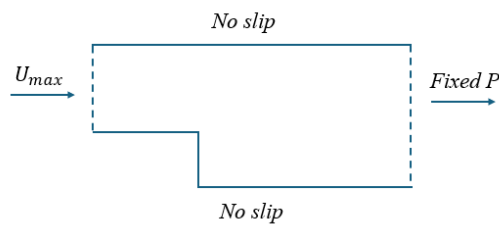
WC6 – Boundary Condition Study for Backward-Facing Step Flow

Objective

The objective of this study is to investigate how different boundary conditions affect the flow behavior in a backward-facing step problem using a finite-volume CFD solver.

The benchmark case is first simulated and then modified boundary conditions are introduced. The effects on the recirculation zone, reattachment length, pressure/velocity distributions, numerical stability, and convergence behavior are compared.

1. Benchmark Case Code Modifications



File changed: bound.m

Original code

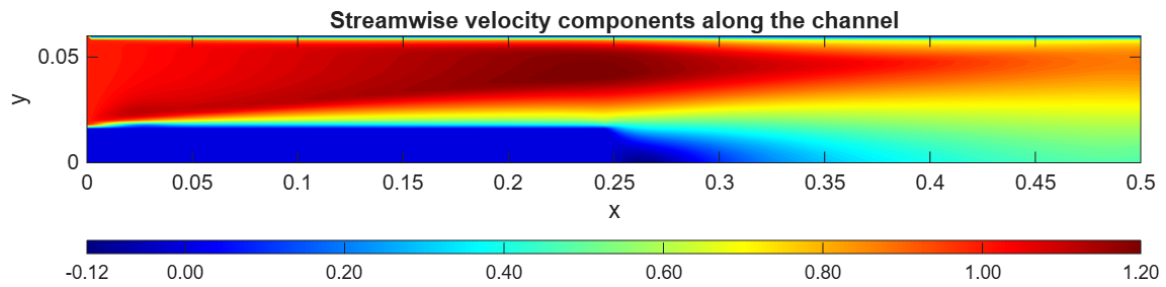
```
for J = NhJ+1:NPJ+2
    u(2,J) = 0; % inlet velocity
    k(1,J) = 1.5*(U_IN*Ti)^2;
    eps(1,J) = Cmu^0.75 *k(1,J).^1.5/(0.07*(YMAX-h)*0.5);
end
```

Modified code

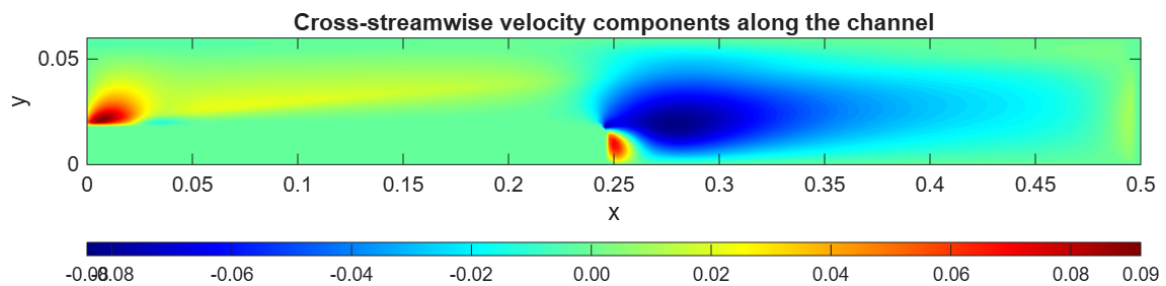
```
for J = NhJ+1:NPJ+2
    u(2,J) = U_IN; % uniform inlet velocity
    v(1,J) = 0; % no vertical inlet velocity
    k(1,J) = 1.5*(U_IN*Ti)^2;
    eps(1,J) = Cmu^0.75 *k(1,J).^1.5/(0.07*(YMAX-h)*0.5);
end
```

2. Benchmark Results

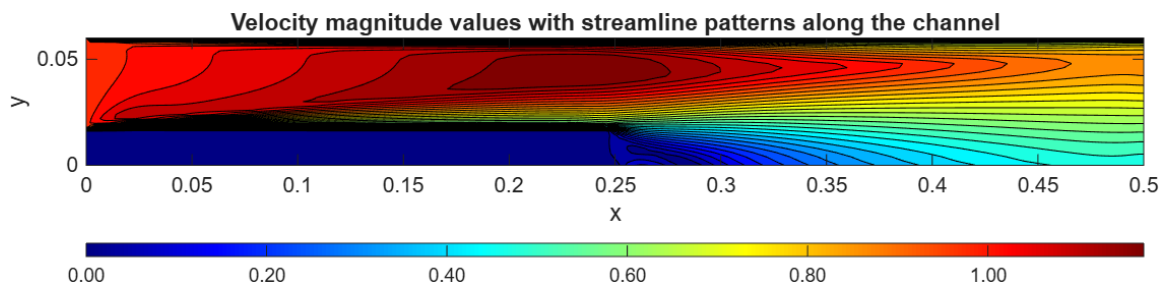
a) Streamwise velocity contour



b) Cross-stream velocity contour



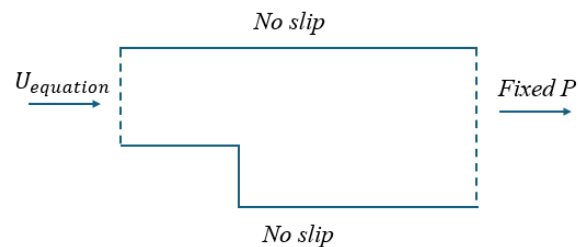
c) Velocity magnitude and streamlines



3. Modified Boundary Conditions

The following boundary conditions were modified to investigate their influence on the flow behavior.

3.1. Parabolic inlet profile



File changed: bound.m

Beware that the given velocity profile is not directly applicable. Check the boundary conditions yourself: $u(h) = u(3h) = 0$.

Derive the velocity profile yourself from the plane Poiseuille equation.

$$\iint \frac{d_u^2}{d_y^2} dy dy = \iint \frac{-1}{\mu} \frac{d_p}{d_x} dy dy$$

The final equation becomes:

$$u(y) = U_{max} \left(\frac{-y^2 + 4hy - 3h^2}{h^2} \right)$$

Modified code

```
for J = NhJ+1:NPJ+2
```

```
    u(2,J) = U_IN*(-y_v(J)^2+4*h*y_v(J)-3*h^2)/h^2; % parabolic inlet
```

```
    v(1,J) = 0;
```

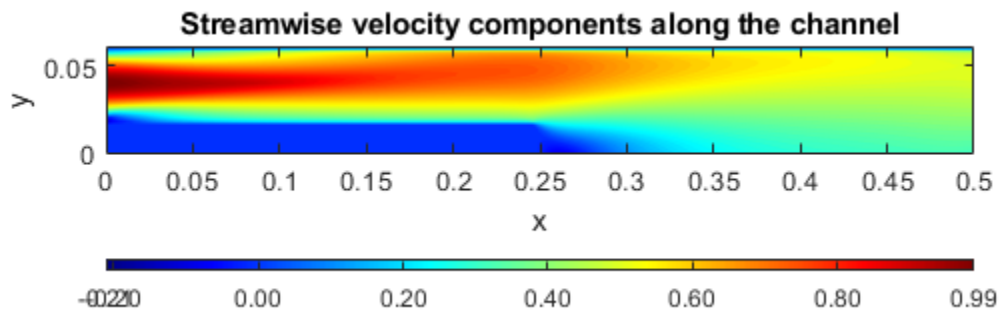
```
    k(1,J) = 1.5*(U_IN*Ti)^2;
```

```
    eps(1,J) = Cmu^0.75*k(1,J).^1.5/(0.07*(YMAX-h)*0.5);
```

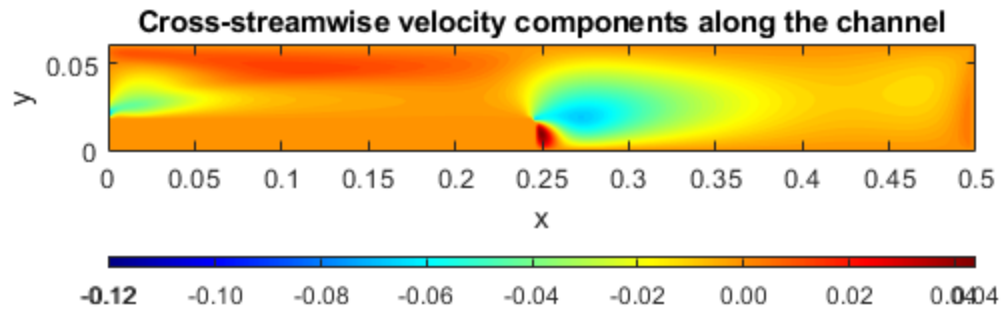
```
end
```

3.1.1. Parabolic inlet profile results

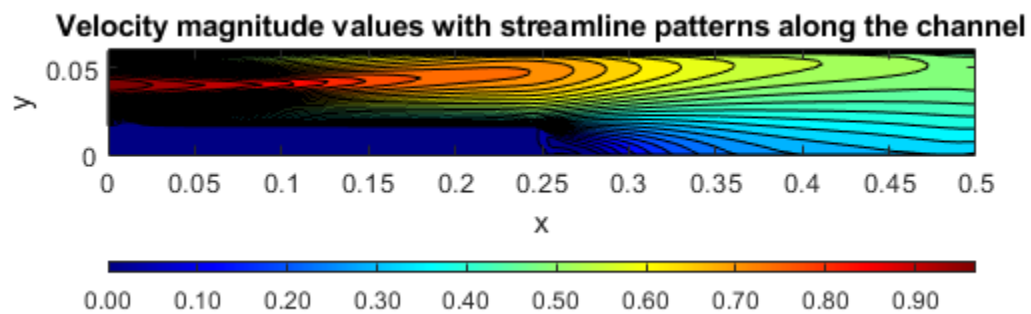
a) Streamwise velocity contour



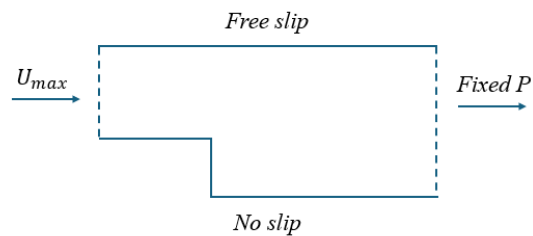
b) Cross-stream velocity contour



c) Velocity magnitude and streamlines



3.2. Slip upper wall



File changed: bound.m

Modified code

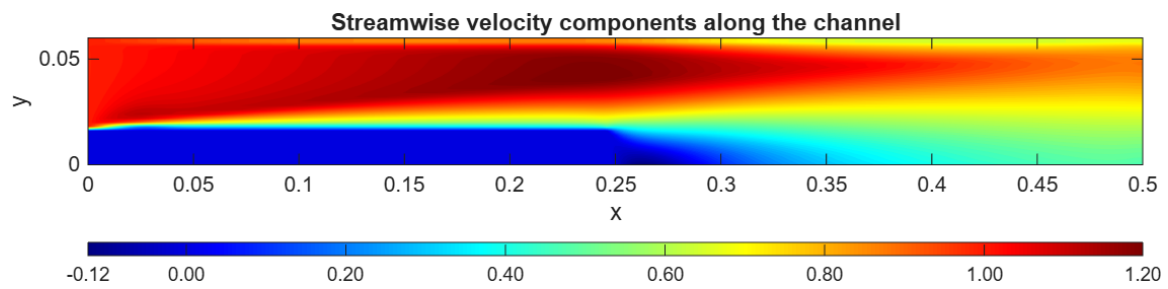
% Case slip wall at top boundary

$u(:,NPJ+2) = u(:,NPJ+1);$ % $du/dy = 0$

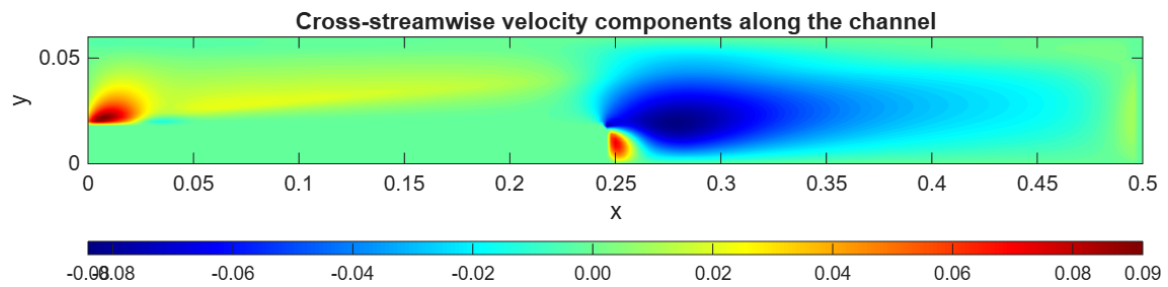
$v(:,NPJ+2) = 0;$ % no penetration

3.2.1. Slip upper wall results

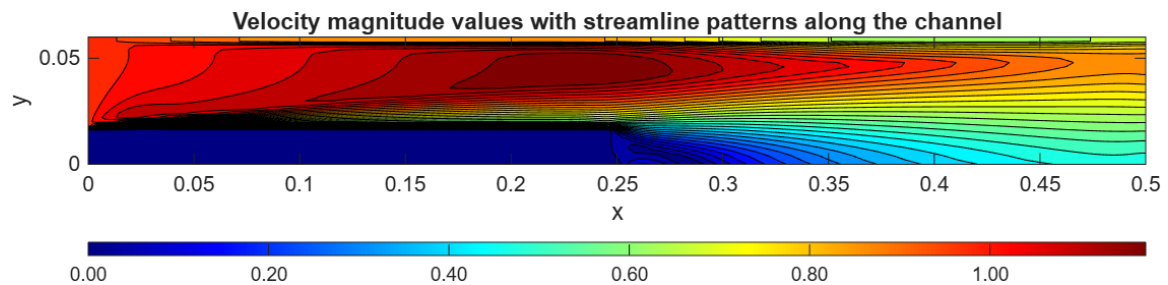
Streamwise velocity contour



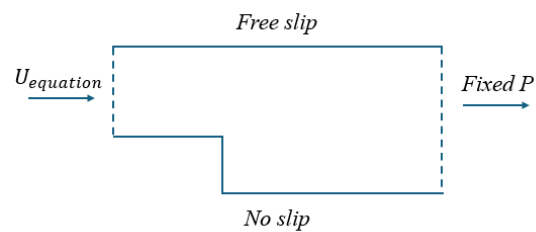
Cross-stream velocity contour



Velocity magnitude and streamlines



3.3 Combined



File changed: bound.m

Modified code

```
for J = NhJ+1:NPJ+2
```

```
    u(2,J) = U_IN*(-y_v(J)^2+4*h*y_v(J)-3*h^2)/h^2; % parabolic inlet
```

```
    v(1,J) = 0;
```

```
    k(1,J) = 1.5*(U_IN*Ti)^2;
```

```
    eps(1,J) = Cmu^0.75*k(1,J).^1.5/(0.07*(YMAX-h)*0.5);
```

```
end
```

Modified code

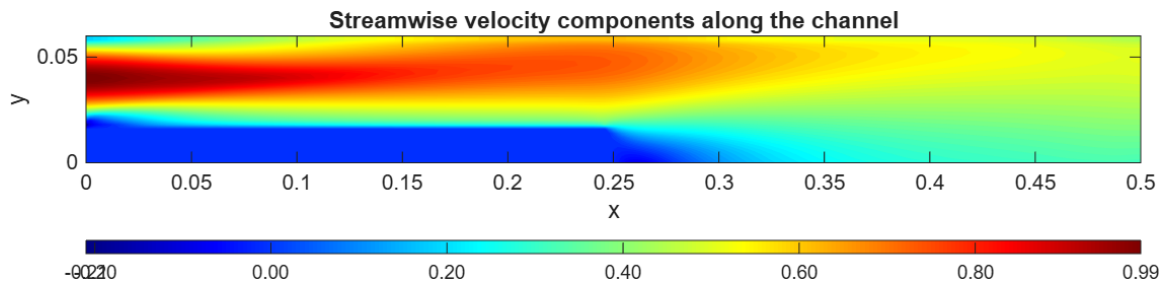
```
% Case slip wall at top boundary
```

```
u(:,NPJ+2) = u(:,NPJ+1); % du/dy = 0
```

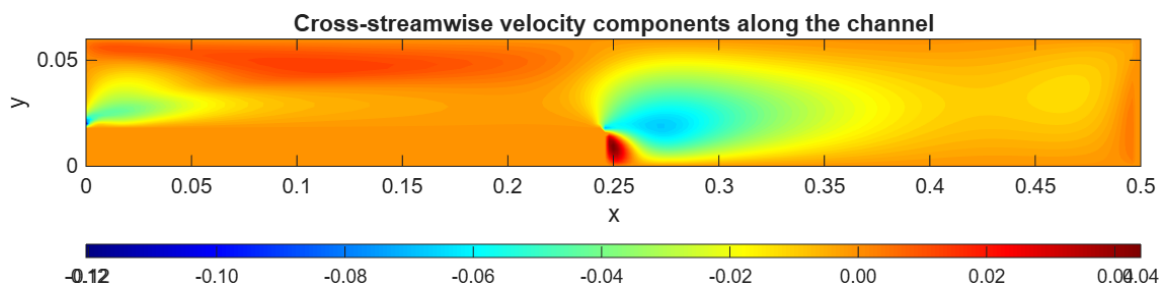
```
v(:,NPJ+2) = 0; % no penetration
```

3.3.1. Combined result

a) Streamwise velocity contour



b) Cross-stream velocity contour



c) Velocity magnitude and streamlines

